A source-centred ridge for heteroscedastic probit models with few target rows

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Abstract

A heteroscedastic probit, a probit whose noise scale depends on features, estimates its scale coefficients poorly from a few hundred target rows. We centre its ridge penalty at a fit on a large related source sample and pick the strength by grouped cross-validation on a grid that holds both endpoints, target-only and source-only. Against the plain ridge-at-zero target fit of the same package (`scaleprobit` 0.1.0), on a public football era-shift task with 852 later test games, log-loss changes by -0.0265 nats per row (SE 0.0077) at 50 target games and by -0.0056 (SE 0.0014) with the full 284-game season; on a men's-to-women's basketball task (838 test games) it ties. Using the source fit unchanged is as good or better: it beats ours by +0.0058 (SE 0.0025) on football at the full season and by +0.0324 (SE 0.0145) on basketball at 25 games, and ties elsewhere, so the gain over target-only comes from the source fit, not from adapting to the target. Our pre-registered test (beat target-only, source-only and pooled at once) failed on both datasets. Differences are paired on the same test rows over 20 target subsamples per size; SEs bootstrap whole test games.

1 Problem

Many binary outcomes are threshold crossings of a quantity that keeps moving. A game ends with the home team ahead or not; a temperature ends above a mark or not. The classic answer is a Brownian closed form [Stern, 1994]: with signed distance d to the threshold, remaining horizon h and base scale s (the typical movement per unit of $\sqrt{h}$), the probability is $\Phi(d/(s\sqrt{h}))$. Its weak spot is the scale. How far the quantity can still move depends on the state of the game, so a heteroscedastic probit [Harvey, 1976] learns the scale as a log-linear function of features, as weather post-processing does [Gneiting et al., 2005, Messner et al., 2014].

We hit the limit of this model when the target sample is small. With a few hundred target rows the scale coefficients are poorly determined, and a plain ridge shrinks them towards zero, which is towards the closed form again. Often a large related sample exists: earlier seasons, another league, another product. Fitting on it alone ignores how the target differs, and pooling needs the source rows at fit time. We ask whether a ridge pulled towards the source fit, instead of towards zero, beats all three.

2 The change

We keep the model fixed and change only the penalty. For row i the fitted scale, the score and the probability are

$$\sigma_i = s_i \exp(a + w^\top z_i), \qquad \eta_i = \frac{u_i}{\sigma_i} + b, \qquad P(y_i = 1) = F\big(\mathrm{clip}(\eta_i, -c, c)\big), \qquad u_i = \frac{d_i}{\sqrt{h_i}}. \tag{1}$$

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With parameters $\theta = (a, w, b)$ set to zero, this is the closed form of Section 1. The plain variant penalises towards zero, and we penalise towards the source fit θ_0:

$$\hat{\theta}_{\text{plain}}(\lambda) = \arg\min_{\theta} \; \ell_T(\theta) + \tfrac{\lambda}{2}\big(\|w\|^2 + b^2\big), \qquad \hat{\theta}(\lambda) = \arg\min_{\theta} \; \ell_T(\theta) + \tfrac{\lambda}{2}\|\theta - \theta_0\|^2. \tag{2}$$

Symbol	Meaning
d_i, h_i, s_i	signed distance to the threshold, remaining horizon, base scale
z_i	log scale features, standardised with the *source* mean and sd
a, w, b	scale intercept, scale slopes, bias of the score
F, c	standard normal CDF (or Student-t), clip bound $c = 12$
ℓ_T	summed negative log-likelihood on the target rows
θ_0	the source fit (plain variant with $\lambda = 1$ on the source)

Three choices make it more than a prior mean. First, the penalty covers every coordinate, including the scale intercept a, which the plain variant leaves free, because we expect the overall level of the scale to be the term a small target estimates worst (the post-hoc ablation in Table 3 gives -0.0008 on football and a tie on basketball). Second, we freeze the standardisation of z at the source values, so θ_0 means the same thing on both samples. Third, λ comes from grouped cross-validation (folds that keep all rows of a game together) over $\{0, 1, 10, 100, 1000, 10000, \infty\}$ with 5 folds, and ties go to the larger λ. Both endpoints are on the grid: $\lambda = \infty$ returns θ_0 exactly (source-only), and $\lambda = 0$ is the unpenalised target fit started at θ_0 (target-only). The CV table therefore shows directly whether adapting to the target paid. We fit with L-BFGS-B [Byrd et al., 1995], a quasi-Newton solver with box bounds, using an analytic gradient and boxes $|a| \le 1.5$, $|w_j|, |b| \le 2$. The gradient is zero on clipped rows, where the loss is flat.

None of this is new in general. A ridge centred at a source fit is MAP adaptation (a prior centred at the old model, Chelba and Acero, 2006), biased regularisation [Kuzborskij and Orabona, 2013] and L2-SP (an L2 penalty towards pretrained weights, Li et al., 2018), and Tian and Feng [2023] study GLM transfer with a CV-chosen strength. Our delta is narrow: the heteroscedastic probit with the three choices above, measured against both endpoints. Proposition 1 restates the classic local argument for this likelihood; the proof is in Appendix A.

Proposition 1 (local risk, after Kuzborskij and Orabona, 2013)**.** *Take the quadratic approximation of ℓ_T around the target optimum θ_*, with total Fisher information (the curvature of the log-likelihood) $J \succ 0$ from n target rows, penalty $\frac{\lambda}{2}(\theta - \theta_0)^\top \Lambda (\theta - \theta_0)$ with Λ diagonal, $\Lambda \succeq 0$, $\Lambda \neq 0$, and gap $\delta = \theta_0 - \theta_*$. The expected excess risk $R(\lambda)$ in the per-row metric J/n has slope $R'(0) = -2\,\mathrm{tr}(J^{-1}\Lambda)/n < 0$. So some $\lambda > 0$ beats target-only for any finite δ, and for $\Lambda \succ 0$, $R(\lambda) \to \delta^\top (J/n)\,\delta$ as $\lambda \to \infty$.*

So in the quadratic regime a small pull always lowers the expected risk. The proposition does not say cross-validation finds the right λ on a few dozen groups, and the likelihood is not convex, so we claim no unique optimum.

Using it on other data. The method needs, per row, a binary outcome y, a signed distance d to the threshold (positive when $y = 1$ is ahead), a horizon $h > 0$ left until the outcome is decided, a base scale $s > 0$ (one constant is enough, for instance the source's spread of final changes per $\sqrt{h}$) and positive scale features x (we use z = the standardised $\log x$). The recipe is three steps: (1) fit the plain variant on the source with $\lambda = 1$ and freeze its parameters and standardisation as θ_0; (2) on the target rows, score the grid above by grouped CV, grouping rows that share an outcome; (3) refit the chosen λ on all target rows and read the CV table, which says whether target-only, source-only or a point in between won. In the package this is `prior = fit_source(Xs, ys, distance=ds, horizon=hs, base_scale=ss)` and then `ScaleProbitTransferCV(prior=prior).fit(Xt, yt, distance=dt, horizon=ht, base_scale=st, groups=gt)`; `cv_results_` holds the table.

3 Setup

Data. Football (play-by-play from Ho and Carl, 2024): source seasons 2009–2017 (2,398 games, 47,960 rows), target pool season 2021 (284 games), test seasons 2022–2024 (852 games, 17,040 rows). In both sports the horizon h is the seconds of regulation left and the base scale s is one

constant, the standard deviation of the source's final margins divided by the square root of a full game's seconds, so the closed form is calibrated on the source alone. On football the distance d is the home margin plus the pregame point spread times the share of the game left (Stern's drift: the favourite's expected remaining edge counts as distance); the 4 scale features are the remaining time, the margin, timeouts left and the pregame total-points line. Basketball [Gilani and Hutchinson, 2024]: source men's league 2019–2024 (7,565 games, 151,300 rows), target pool women's league 2019–2022 (817 games), test 2023–2025 (838 games, 16,760 rows); d is the home margin and the 3 scale features are the remaining time, the scoring rate so far and the margin. Each game gives 20 random game states, and $y = 1$ when the home team wins. The test seasons come after every training season, so the football task is an era shift (train on old seasons, test on new ones), and nothing tunes on them. A synthetic generator with known truth gives a regime map over 20 cells (shift $\times$ target rows, 20 draws each); the shift is the Euclidean length of the difference between the target's and the source's true θ.

Arms. Closed form ($\theta = 0$, the trivial baseline); target-only (the package's unmodified plain variant at the same version, λ by the same CV; this is the upstream the change is measured against); source-only (θ_0 as it is); pooled (source and target rows with a domain indicator in the scale); ours. Generic learners: logistic regression [Pedregosa et al., 2011] target-only and pooled, on $[u, uz, z]$ (the linearised scale model); LightGBM (gradient-boosted trees, Ke et al., 2017) target-only, source-only and fine-tuned from the source booster, on the raw state variables (d, h, margin, lines, timeouts or scoring rate, u). 5 pre-registered ablations: λ fixed at 1, 100 and 10000, a Student-$t(4)$ link, and target standardisation. Every arm sees the same target rows and the same source rows where it uses any; each picks its one capacity knob by the same 5-fold grouped CV on the target rows (λ, the logistic penalty, or LightGBM's boosting rounds).

Metric and SE. The primary metric is held-out log-loss $L = -\frac{1}{N}\sum_i [y_i \log p_i + (1-y_i)\log(1-p_i)]$ in nats per row. Target sizes are 25, 50, 100, 200 (and 400 for basketball) games, 20 random subsamples each, plus the full pool. The paired SE of $L_{\text{ours}} - L_{\text{arm}}$ is a block bootstrap over test games (2,000 draws), resampling whole games because rows of one game are dependent, plus the between-subsample variance divided by 20. We call a difference inside 1.96 SE a tie. Before the run we pre-registered the primary test: ours against target-only, source-only and pooled at every subsampled size, with success if, at some subsampled size, all three paired upper 95% bounds are below zero on either real dataset.

4 Results

Ours beats target-only on football and does not beat source-only on either real dataset. Table 1 gives the full-season fit, where cross-validation picked $\lambda = 1000$ for ours on both datasets, so the fit moved away from the source.

Table 1: Ours beats the target-only fit on football and loses to source-only and pooling there; on basketball every scaleprobit arm ties. Full target season, held-out log-loss (nats per row, lower is better) and the paired difference ours − arm with its SE; negative means ours is better.

	Football (284 target, 852 test games)		Basketball (817 target, 838 test games)	
Arm	log-loss	ours − arm (SE)	log-loss	ours − arm (SE)
Closed form	0.4908	−0.0202 (0.0054)	0.4732	−0.0016 (0.0028)
Target-only	0.4762	−0.0056 (0.0014)	0.4719	−0.0004 (0.0006)
Source-only	0.4648	+0.0058 (0.0025)	0.4709	+0.0006 (0.0022)
Pooled	0.4652	+0.0054 (0.0021)	0.4716	−0.0000 (0.0007)
Ours	**0.4706**		**0.4715**	
Logistic reg., target	0.4751	−0.0045 (0.0016)	0.4715	+0.0000 (0.0004)
Logistic reg., pooled	0.4678	+0.0028 (0.0021)	0.4711	+0.0005 (0.0018)
LightGBM, target	0.5094	−0.0388 (0.0076)	0.4782	−0.0067 (0.0012)
LightGBM, source	0.4628	+0.0078 (0.0046)	0.4699	+0.0016 (0.0023)
LightGBM, fine-tuned	0.4760	−0.0055 (0.0055)	0.4705	+0.0010 (0.0014)

On football ours lowers the target-only log-loss from 0.4762 to 0.4706, a paired gain of −0.0056 (SE 0.0014). But source-only scores 0.4648 and pooling 0.4652, both better than ours beyond their

SE. Our reading: the finite λ fits quirks of the one target season that the test seasons do not share. LightGBM fitted on the source alone has the lowest log-loss on both datasets (0.4628 and 0.4699); ours never beats it and loses to it beyond SE at 100 football games and 25 and 50 basketball games. On basketball the women's league is close to Brownian: the closed form alone scores 0.4732 against 0.4715 for ours, so the scale has little left to learn; at 25 basketball games ours is even worse than the closed form (+0.0301, SE 0.0141). Figure 1 gives every target size (numbers in Table 2).

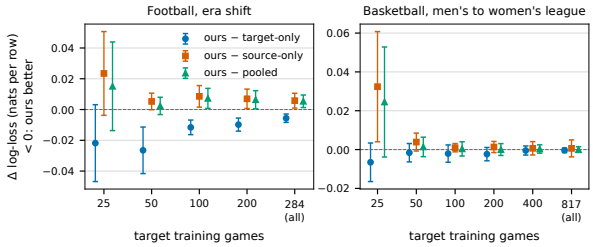


Figure 1: Ours beats target-only on football from 50 games on, but source-only and pooling sit at or below it from 50 games on; on basketball everything ties except the 25-game loss to source-only. Paired held-out log-loss difference ours − rival, bars are 95% intervals (± 1.96 SE).

The largest football gain over target-only is -0.0265 (SE 0.0077) at 50 games; at 25 games the mean gain -0.0218 is within 1.96 SE (SE 0.0127), a tie. Against the uncorrected source fit the sign never flips: from 100 games on that endpoint wins beyond SE, and at 50 games it is just inside the tie band (+0.0053, SE 0.0027). The worst basketball case is the smallest target, where the centred ridge trails source-only by +0.0324 (SE 0.0145), because CV on 25 games picked $\lambda = \infty$ in only 8 of 20 subsamples and $\lambda = 0$ in 1.

Calibration tells the same story (Figure 2): on football the expected calibration error (ECE, Guo et al., 2017, the mean gap between forecast and observed win rates over 15 equal-width bins) is 0.0326 for target-only, 0.0297 for ours and 0.0155 for source-only.

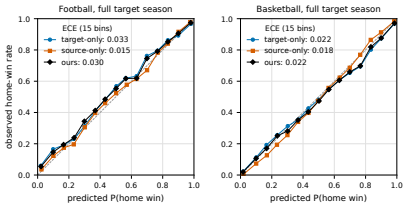


Figure 2: Source-only is the best calibrated arm on both datasets; ours sits between it and target-only. Reliability diagrams at the full target season, bins with at least 30 rows, ECE in the legend.

Synthetic data, where the true shift is known, shows when adaptation pays (Figure 3 in Appendix B). The centred ridge improves on target-only beyond SE in 15 of 20 cells and never loses to it. It overtakes source-only in 5 cells, all at shift 0.3 or 0.6 with enough target rows, and falls behind in 4, where the shift is small. All three rivals fall at once in 1 cell (shift 0.3, 800 rows: -0.0012, -0.0091, -0.0077). Which endpoint wins depends on the shift, and on a small target grouped CV does not reliably tell a real shift from noise.

5 Limits

The pre-registered success test failed on both real datasets, at every size. The method improves on the plain target fit on football from 50 games on and in 15 of 20 synthetic cells, and ties it on basketball; on both real tasks the best single choice was to use the source fit as it is, so the gain over target-only comes from the source fit, not from target adaptation. We do not claim the method beats source-only or pooling, nor that it generalises beyond these two sports and the synthetic generator.

After the main run we tried 4 post-hoc variants of the λ selection (a 13-point grid, slopes pulled harder, the one-SE rule, repeated CV; Appendix B); none was kept, and the one-SE rule is source-only by another name. The 5 ablations in Table 3 were pre-registered; its free-intercept row is post-hoc.

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A Proof of Proposition 1

Given. 1) Quadratic approximation: $\ell_T(\theta) \approx \ell_T(\hat\theta) + \frac{1}{2}(\theta-\hat\theta)^\top J(\theta-\hat\theta)$ with $J \succ 0$ and $\hat\theta$ the unpenalised target fit. 2) $J(\hat\theta - \theta_*) = g$ with $\mathbb{E}[g] = 0$ and $\mathrm{Cov}(g) = J$. 3) Objective $Q_\lambda(\theta) = \ell_T(\theta) + \frac{\lambda}{2}(\theta-\theta_0)^\top\Lambda(\theta-\theta_0)$, Λ diagonal, $\Lambda_{jj} \geq 0$, at least one $\Lambda_{jj} > 0$ (the package uses $\Lambda = I$). 4) Risk $R(\lambda) = \mathbb{E}\big[(\hat\theta(\lambda)-\theta_*)^\top(J/n)(\hat\theta(\lambda)-\theta_*)\big]$. 5) Matrix rules: $\frac{d}{d\lambda}A^{-1} = -A^{-1}\frac{dA}{d\lambda}A^{-1}$; $\mathrm{tr}(XY) = \mathrm{tr}(YX)$; $\mathbb{E}[g^\top Mg] = \mathrm{tr}(MJ)$ for $\mathbb{E}[g] = 0$, $\mathrm{Cov}(g) = J$.

Choose: $A = J + \lambda\Lambda$ and $m = \lambda A^{-1}\Lambda\delta$.

Need to prove: $\hat\theta(\lambda) - \theta_* = m + A^{-1}g$.

$$
\begin{aligned}
0 &= J(\theta-\hat\theta) + \lambda\Lambda(\theta-\theta_0) && \text{gradient of } Q_\lambda\text{, given 1 and 3}\\
A\theta &= J\hat\theta + \lambda\Lambda\theta_0 && \text{collect the terms in } \theta\\
A(\theta-\theta_*) &= J(\hat\theta-\theta_*) + \lambda\Lambda(\theta_0-\theta_*) && \text{subtract } A\theta_* = J\theta_* + \lambda\Lambda\theta_*\\
\theta-\theta_* &= A^{-1}g + \lambda A^{-1}\Lambda\delta && \text{given 2, multiply by } A^{-1}
\end{aligned}
$$

The last line needs A invertible, therefore, for $x \neq 0$:

$$
\begin{aligned}
x^\top Ax &= x^\top Jx + \lambda\, x^\top\Lambda x && \text{definition of } A\\
&> 0 && J \succ 0,\ \lambda \geq 0,\ \Lambda \succeq 0
\end{aligned}
$$

Same method, on the risk. Substitution in given 4:

$$
\begin{aligned}
R(\lambda) &= \mathbb{E}\big[(m + A^{-1}g)^\top\tfrac{J}{n}(m + A^{-1}g)\big] && \text{first part}\\
&= m^\top\tfrac{J}{n}m + 2\,m^\top\tfrac{J}{n}A^{-1}\mathbb{E}[g] + \mathbb{E}\big[g^\top A^{-1}\tfrac{J}{n}A^{-1}g\big] && \text{expand, } m \text{ is not random}\\
&= \underbrace{\lambda^2\delta^\top\Lambda A^{-1}\tfrac{J}{n}A^{-1}\Lambda\delta}_{B(\lambda)} + \underbrace{\tfrac{1}{n}\mathrm{tr}(A^{-1}JA^{-1}J)}_{V(\lambda)} && \mathbb{E}[g] = 0\text{, given 5}
\end{aligned}
$$

Need to prove: $R'(0) = -2\,\mathrm{tr}(J^{-1}\Lambda)/n$. Choose $\beta(\lambda) = \delta^\top\Lambda A^{-1}\frac{J}{n}A^{-1}\Lambda\delta$, so $B(\lambda) = \lambda^2\beta(\lambda)$, with β differentiable at 0 because A^{-1} is (given 5, A invertible above).

$$
\begin{aligned}
B'(\lambda) &= 2\lambda\,\beta(\lambda) + \lambda^2\beta'(\lambda) && \text{product rule}\\
B'(0) &= 2\cdot 0\cdot\beta(0) + 0\cdot\beta'(0) = 0 && \lambda = 0
\end{aligned}
$$

For the variance:

$$
\begin{aligned}
V'(\lambda) &= \tfrac{1}{n}\big[\mathrm{tr}(-A^{-1}\Lambda A^{-1}JA^{-1}J) + \mathrm{tr}(A^{-1}J(-A^{-1}\Lambda A^{-1})J)\big] && \text{product rule, given 5}\\
V'(0) &= -\tfrac{1}{n}\big[\mathrm{tr}(J^{-1}\Lambda J^{-1}JJ^{-1}J) + \mathrm{tr}(J^{-1}JJ^{-1}\Lambda J^{-1}J)\big] && A = J \text{ at } \lambda = 0\\
&= -\tfrac{1}{n}\big[\mathrm{tr}(J^{-1}\Lambda) + \mathrm{tr}(J^{-1}\Lambda J^{-1}J)\big] && J^{-1}J = I\\
&= -\tfrac{2}{n}\mathrm{tr}(J^{-1}\Lambda) && \text{cyclic trace, given 5}
\end{aligned}
$$

The sign needs $\operatorname{tr}(J^{-1}\Lambda) > 0$, therefore:

$$\begin{aligned} \operatorname{tr}(J^{-1}\Lambda) &= \textstyle\sum_j (J^{-1})_{jj}\Lambda_{jj} && \Lambda \text{ diagonal} \\ &= \textstyle\sum_j (e_j^\top J^{-1} e_j)\,\Lambda_{jj} && \text{unit vectors } e_j \\ &> 0 && J^{-1} \succ 0, \text{ all } \Lambda_{jj} \geq 0, \text{ one} > 0 \end{aligned}$$

Therefore $R'(0) = B'(0) + V'(0) = -2\operatorname{tr}(J^{-1}\Lambda)/n < 0$, and some $\lambda > 0$ has $R(\lambda) < R(0)$, the target-only risk.

Same method, on the limit ($\Lambda \succ 0$).

$$\begin{aligned} \lambda A^{-1} = (J/\lambda + \Lambda)^{-1} &\to \Lambda^{-1} && \text{divide } A \text{ by } \lambda \\ m = \lambda A^{-1}\Lambda\delta &\to \delta, \qquad A^{-1} \to 0 && \text{first line} \\ R(\lambda) &\to \delta^\top \tfrac{J}{n}\delta + 0 && \text{substitution in } B \text{ and } V \end{aligned}$$

That is the source-only risk. □

B Learning curve, ablations and post-hoc variants

Table 2 gives the numbers behind Figure 1, and Figure 3 is the synthetic regime map of Section 4.

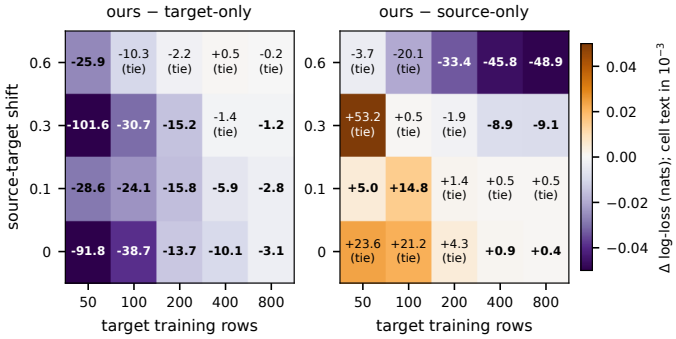


Figure 3: Ours beats target-only in 15 of 20 cells but beats source-only only when the shift is large and the target has enough rows. Synthetic regime map, paired log-loss difference over 20 draws; plain text marks a tie.

Table 2: No size meets the pre-registered test: ours never beats source-only, and only beats target-only on football. Paired held-out log-loss difference ours − rival (SE) per target size, mean over 20 target subsamples; **bold** is a win beyond 1.96 SE.

target games	ours − target-only	ours − source-only	ours − pooled
Football			
25	−0.0218 (0.0127)	+0.0234 (0.0139)	+0.0151 (0.0147)
50	**−0.0265 (0.0077)**	+0.0053 (0.0027)	+0.0023 (0.0029)
100	**−0.0116 (0.0024)**	+0.0086 (0.0036)	+0.0073 (0.0033)
200	**−0.0098 (0.0022)**	+0.0070 (0.0032)	+0.0064 (0.0030)
all 284	**−0.0056 (0.0014)**	+0.0058 (0.0025)	+0.0054 (0.0021)
Basketball			
25	−0.0065 (0.0051)	+0.0324 (0.0145)	+0.0245 (0.0144)
50	−0.0016 (0.0024)	+0.0039 (0.0024)	+0.0014 (0.0026)
100	−0.0021 (0.0023)	+0.0010 (0.0011)	+0.0004 (0.0019)
200	−0.0023 (0.0018)	+0.0013 (0.0015)	+0.0000 (0.0016)
400	−0.0005 (0.0012)	+0.0007 (0.0018)	+0.0003 (0.0011)
all 817	−0.0004 (0.0006)	+0.0006 (0.0022)	−0.0000 (0.0007)

Table 3 removes one design choice at a time at the full season (each of the three choices of Section 2 has a row: λ by CV, source standardisation, the penalised intercept), and Table 4 compares the 4 post-hoc selection variants with the default on the same target subsamples.

Table 3: Cross-validated λ beats fixed small λ on football but loses to the fixed large one, which sits next to source-only; source standardisation helps on football, and penalising the scale intercept is worth -0.0008 there (about 2 SE). Full target season, ours $-$ ablation (SE); the free-intercept row is a paired bootstrap over test games of one refit.

Ablation	Football	Basketball
$\lambda = 1$ fixed	-0.0110 (0.0025)	-0.0011 (0.0006)
$\lambda = 100$ fixed	-0.0084 (0.0020)	-0.0009 (0.0005)
$\lambda = 10000$ fixed	+0.0051 (0.0019)	+0.0013 (0.0014)
Student-$t(4)$ link	+0.0012 (0.0011)	-0.0001 (0.0005)
Target standardisation	-0.0014 (0.0006)	+0.0003 (0.0002)
Scale intercept a free (post-hoc)	-0.0008 (0.0004)	+0.0000 (0.0008)

Table 4: No post-hoc variant beats the default by enough to keep; the one-SE rule wins beyond SE at two sizes (100 football games, 25 basketball games) only because it always returns source-only. Variant $-$ default (SE) on the same subsamples.

Variant	25 games	50 games	100 games
Football			
13-point grid	+0.0003 (0.0093)	+0.0011 (0.0018)	-0.0019 (0.0018)
Slopes anchored	+0.0214 (0.0170)	-0.0012 (0.0030)	-0.0004 (0.0030)
One-SE rule	-0.0234 (0.0139)	-0.0053 (0.0027)	-0.0086 (0.0036)
Repeated CV	-0.0183 (0.0134)	+0.0071 (0.0073)	-0.0043 (0.0024)
Basketball			
13-point grid	+0.0057 (0.0054)	-0.0004 (0.0015)	+0.0007 (0.0006)
Slopes anchored	-0.0078 (0.0068)	+0.0010 (0.0033)	+0.0013 (0.0011)
One-SE rule	-0.0324 (0.0145)	-0.0039 (0.0024)	-0.0010 (0.0011)
Repeated CV	-0.0120 (0.0094)	+0.0007 (0.0024)	-0.0002 (0.0008)

C Origin, reproduction and cuts

Origin. We first built the method for a private application, and this is the one statement we make about it. On a proprietary binary forecasting task (a few hundred target training examples; a related source sample more than an order of magnitude larger), centring the ridge at the source fit lowered held-out log-loss by 0.0056 nats (0.3161 $\rightarrow$ 0.3105) and ECE from 0.030 to 0.027 over the target-only fit chosen by an extensive cross-validated search (paired block-bootstrap SE of the log-loss delta 0.0032, so about 1.7 SE and a tie under the 1.96 SE rule used here; the 95 % interval, -0.0120 to +0.0007, includes 0). Cross-validation chose a lambda so large that the fit stayed within 0.003 of the source parameters: at this target size the gain came from the source fit, and target adaptation added nothing measurable. Against an earlier, lightly tuned target-only configuration the difference was within noise (paired SE 0.0005-0.0009).

Reproduction. The package is `scaleprobit` (Python, SciPy L-BFGS-B, Virtanen et al., 2020). Benchmark results are those committed at `3e8d758` with package version 0.1.0; the package is our own, not a vendored library, so there is no upstream diff, and the plain variant in the same package is the unmodified baseline. For scale only (one run, one multi-core CPU machine, not a speed claim), the benchmark took 856 s (football), 1,376 s (basketball) and 2,628 s (synthetic) of wall time. Commands, from the repository root: `python -m bench.run_football`, `bench.run_hoops`, `bench.run_synthetic`, `bench.variants.run football|hoops`, `bench.variants.synthetic_one_se`; then, with `PYTHONPATH=src:.`, `paper/results/refit_full.py`, `paper/results/ablate_free_intercept.py`, `paper/results/make_numbers.py` and `paper/figures/make.py` regenerate every number and figure in this note. The full-season refit reproduces the benchmark log-loss of every scaleprobit arm, and its per-game SE matches the block SE at the full season (0.0014 against 0.0014 on football). Cuts, none silent: a third real dataset; the R arms `glmx::hetglm` [Zeileis et al., 2026] and `glmtrans`; the Platt and grouped-λ ablations, and the free-intercept ablation at subsampled sizes (it was run afterwards at the full season only); the football source starts later than planned to bound download

and fit time. Basketball target files carry no usable pregame spread, so neither side uses Stern's drift there. LightGBM uses one fixed parameter set; only its rounds are chosen by grouped CV.